

Week 6: Final Exam

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1. The two methods are p-values and correlation analysis. P-values can help determine if a variable is having a statistically significant impact and correlation can highlight variables that track closely with the target. The final feature set can vary depending on which method you use. This is because a variable may appear strong by just looking at correlation, but can turn into a weak variable when other predictors are added to the regression.
2. Because R-squared usually increases when more variables are added, even when those new variables add little real value. AIC is helpful because it rewards model fit but also penalizes unnecessary complexity, so it gives a better sense of whether the model is improving in a meaningful way.
3. One way is impute the missing value with median, mean, mode or group by mean/median. Another way is remove the column directly. Both ways should be used in different situations base on business purpose of the variable, and whether it is important to the analysis.
4. This is the question we talked about in class. The company would usually want to reduce false positives which model predicts that the borrower will pay, but they actually default. This bad debt could create direct financial risk for the bank.
5. Logistic regression could be used by a bank to explain how income, debt, and spending behavior affect probability of default. An insurer could use a decision tree to explain a simple set of rules that determine which type of applicants are more likely to file claims. Random forest and gradient boosting are both better for optimization since they allow for more complex relationships to be included in the model, and typically have better predictive performance despite being more difficult to explain to non-technical stakeholders.
6. The review checked the raw datatype, the amount of missing data, distribution shape for numeric fields, category balance for categorical fields, possible outliers, scale differences, and whether any fields were stored in an analytically incorrect type.

Two numeric variables had missing values less than 10%, which are CREDIT_SCORE and ANNUAL_MILEAGE . CREDIT_SCORE showed a normal distribution with slight left skew and some extreme values at the lower and upper ends, while ANNUAL_MILEAGE was closer to symmetric but still contained a wide range of values. Because median is more robust to potential outliers, so both were imputed with median.

	Variable	Mean	Median	Std	Skew	IQR_Outliers	Outlier_Pct	Min	Max
0	CREDIT_SCORE	0.516	0.525	0.138	-0.226	9	0.10	0.053	0.961
1	ANNUAL_MILEAGE	11697.003	12000.000	2818.435	0.040	17	0.19	2000.000	22000.000

SPEEDING_VIOLATIONS, DUIS, and PAST_ACCIDENTS have outliers, they all have high-end values, but all seem reasonable, so I did not make any changes.

POSTAL_CODE was converted from integer to string because it represents a location category rather than a continuous number.

And finally, ID was removed because it is only an identifier.

7. Before Provide the significant variables table, I have to convert categorical variables to dummy variables. However, I found the age and driving experience overlap too strongly in this dataset because a 25 years old drive is impossible to have more than 9 years driver experience. As a result, to avoid multicollinearity I remove the AGE.

From the table below we can see, the most intriguing variable is POSTAL_CODE. This importance implies that perhaps place plays a role in crash risk, which may be a representation of variance in traffic flow, road design, congestion, or driver behavior within the locality as opposed to between drivers.

Variable	Coefficient	p_value
VEHICLE_OWNERSHIP	-1.7304	3.6987283511902034e-143
DRIVING_EXPERIENCE_10-19y	-1.9049	1.930314480182343e-114
DRIVING_EXPERIENCE_20-29y	-3.8815	5.139142496632968e-107
GENDER_male	0.9901	1.3549833621675256e-52
POSTAL_CODE_32765	1.1138	1.66452334883102e-44
DRIVING_EXPERIENCE_30y+	-4.7311	6.996295370647838e-38
POSTAL_CODE_92101	1.3452	4.2094241137073967e-22
ANNUAL_MILEAGE	0.0001	1.6133929613320942e-12
MARRIED	-0.342	1.2179489076525447e-06
INCOME_upper class	-0.3525	0.00012255069877921496
INCOME_poverty	0.4115	0.0002247194525538161
INCOME_working class	0.2132	0.02314952629233038

8. Three models are logistic regression, random forest with 1000 trees, and gradient boosting with 1000 stages. And logistic regression has the highest accurate and recall value overall that make me surprised.

Model	Accuracy	Precision	Recall	F1
0 Logistic Regression	0.8320	0.7454	0.7049	0.7246
1 Random Forest (1000 trees)	0.8005	0.6919	0.6555	0.6732
2 Gradient Boosting (1000 stages)	0.8170	0.7122	0.6986	0.7053

So, the optimized ensemble models did not improve overall accuracy. The reason I think is because most of the predictive relationships in the dataset were fairly stable, and that they could already be modeled to reasonable effect by the linear classifier. The added complexity of the ensemble models did not result in improved generalization to the test set. Their poorer

performance suggests that there wasn't enough excess nonlinear structure in the dataset to benefit from more complicated approaches.

		3 rows × 5 cols			
	Model	TN	FP	FN	TP
0	Logistic Regression	1222	151	185	442
1	Random Forest	1190	183	216	411
2	Gradient Boosting	1196	177	189	438

9.

The insurance company would care most about decreasing false negatives. A false negative is when the model predicts the customer will not have an accident when they actually do. That mistake would cause underwriting to underestimate risk and price the policy too aggressively.

Strictly looking at false negatives, logistic regression had better performance than gradient boosting in this final run with 185 versus 189. Random forest had the highest number of false negatives. Logistic regression was best in both overall accuracy and minimizing the insurance company's most costly error.

10. It looks like accident risk comes less from individual variables and more from clusters of experience, exposure, and behavior. The most dominant driver of risk was driving experience. The more years someone has been driving, the less likely they are to have an accident. This is intuitive since experienced drivers have been exposed to more traffic scenarios and should make steadier decisions. Annual mileage was also a key factor because it serves as a proxy for exposure. The more miles you drive, the more you are exposing yourself to potential accidents. In a way, this model isn't just predicting who will be involved in an accident, but under what conditions they are likely to be involved in one.

One other thing I found interesting is that certain variables indicate behavioral risk while others indicate background risk. For instance, speeding violations and prior accidents can be used to infer whether someone drives safely. Postal code, on the other hand, indicates that the local area can play a role in accident likelihood. This suggests that the probability of an accident is influenced not only by who you are but where and how often you drive. I thought this was useful because it reinforces the idea that insurance risk is derived from both a driver's behavior and the nature of their driving situation.

What I think the insurance company could really benefit from is the idea that applicants should not be separated into high-risk and low-risk groups. There are smaller risk profiles that they should be concerned about. Some applicants are high-risk due to inexperience. Some are high-risk due to high annual mileage. Some applicants have behavioral red flags such as prior accidents or speeding tickets. I think breaking down the model results this way is more practical because the company can apply more nuance to how they price these policies and where they choose to invest effort in intervention.

Top Random Forest feature importances	
10 rows ▾ 10 rows × 1 cols	
	Importance
CREDIT_SCORE	0.226287
ANNUAL_MILEAGE	0.115795
VEHICLE_OWNERSHIP	0.093359
SPEEDING_VIOLATIONS	0.089993
PAST_ACCIDENTS	0.080659
DRIVING_EXPERIENCE_10-19y	0.048354
INCOME_poverty	0.035695
GENDER_male	0.034809
INCOME_upper class	0.034679
DRIVING_EXPERIENCE_20-29y	0.033997

Top logistic coefficients by absolute value	
10 rows ▾ 10 rows × 1 cols	
	Coefficient
POSTAL_CODE_21217	6.285613
DRIVING_EXPERIENCE_30y+	-3.867858
DRIVING_EXPERIENCE_20-29y	-3.625830
DRIVING_EXPERIENCE_10-19y	-1.764248
POSTAL_CODE_92101	1.124390
POSTAL_CODE_32765	1.075301
GENDER_male	0.967857
VEHICLE_OWNERSHIP	-0.781759
INCOME_poverty	0.400811
INCOME_upper class	-0.335493

Top Gradient Boosting feature importances

10 rows ▾ 10 rows × 1 cols

	Importance
SPEEDING_VIOLATIONS	0.177058
VEHICLE_OWNERSHIP	0.172175
CREDIT_SCORE	0.154634
DRIVING_EXPERIENCE_10-19y	0.103629
PAST_ACCIDENTS	0.094879
POSTAL_CODE_21217	0.047671
GENDER_male	0.041691
INCOME_upper class	0.038111
DRIVING_EXPERIENCE_20-29y	0.034032
INCOME_poverty	0.032410